ECG R-Pulse Detector

Multi-stage amplifier for heartbeat signals

ELC1408: Analog Electronics

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by

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Preface

After numerous experiments and attempts over the previous four weeks, on May 13, 2025, I finally managed to design an analog electronic system capable of capturing the ECG signal by capturing its largest and clearest signal, the R wave.

I learned a lot about signals and noise (especially hum noise). I also had to learn some new things like LTspice software. Also, due to the endless problems of this project, I understood many things that were previously unclear, such as how transistors work and the effect of resistor values on the Q-point.

I decided to work on this project alone because difficult projects (for us first-year students) require a clear plan and the absence of disagreements or changes in the plan mid-project. They also require a recognized leader within the team whose final say is decisive (this is difficult to achieve if no student on the team has privileges over the rest of the project).

So, working alone made me the leader, implementer, funder, researcher, and the primary and final decision-maker of the project. Because I worked on this project alone, I initially decided to take responsibility for the project's failure if it failed.

However, thanks to perseverance, the idea didn't fail. Not caring about the project's success or failure, I was able to make the idea a success at the last minute of submitting the project.

Contents

1

Introduction

Electrocardiography, or ECG for short, is one of the biomedical signals used to indicate cardiac activity. The ECG signal differs from other biometric signals, such as EEG and EMG, in that it is easier to monitor because it has a consistent, repetitive pattern and is higher in amplitude than other biometric signals. Therefore, it is suitable for a simple project that does not require an op amp or instrumentation amplifier.

The ECG signal contains five main waves (P, Q, R, S, and T), as shown in Figure 1.1. Each wave represents a cardiac condition. The largest, most obvious, and most important for this project is the R wave, which represents the contraction of the ventricles.

Because each ECG wave contains an R wave, I focused on this wave and decided to ignore the other smaller waves, as they are indicators of the other smaller waves.

I used three ECG/EEG electrodes to capture the signal from the body. Two were located on the right and left wrists to capture the signal, and the third was connected to the ground of the circuit at the elbow so that the body signal would be the same as the signal from the heart. The ground is used to prevent the input signals from becoming floating points. (You can eliminate the need for a third electrode by touching the ground of the circuit with my hand.)

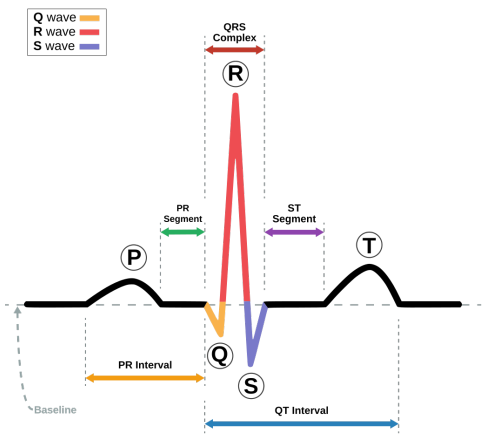

Figure 1.1: ECG signal

2 System description

This analog system was built to be able to do four main things which are the point of this project:

- Ability to capture weak signals (0.1 to 1.5 millivolts), which is the range of the ECG signal.
- Ability to remove non-vital signals from various external noise sources (especially power line noise and hum noise).
- Ability to distinguish the R wave from other waves in the ECG signal.
- Ability to demonstrate the R signal entering the system via a buzzer.

by using multi stage amplifiers and it is not permitted to use:

1. operational amplifiers.
2. power amplifiers.
3. integrated circuits (ICs).
4. board microcontrollers (like Arduino).

To achieve the above, I had to study the ECG wave and analyze it in the time domain and frequency domain and take the results of these analyses in order to design the system as an electrical circuit.

i used a real 10 sec [Figure 2.1] ECG signal from a medical signals archive i found on the internet for a normal person that have no issues in his heart. and used this txt ECG signal file as a reference signal to work on, i used LTspice to do some analyses on the signal and Proteus 8 Professional to implement the system as an electronic circuit.

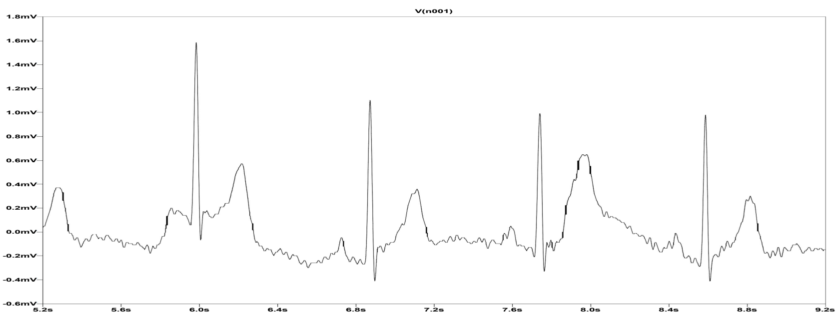

Figure 2.1: 5-second sample of the ECG signal file

As we can see from the signal, it is very weak (in millivolts). Also, the entire signal, with all its components, lies in a frequency range of less than 100 Hz, which is a small range with relatively small frequencies. Therefore, the circuit must have a large input resistance to capture these small frequencies. It must also be able to receive small frequencies of 10 Hz at most.
It must also be able to capture only the R wave, and this can be done in two ways:

1. using a bandpass filter (it must be super sharp)
2. using a clipper (because the R pulse has the greater amplitude)

The R wave does not have a fixed frequency at a specific point, and since it is difficult to design a filter with a very sharp frequency cutoff, I decided to follow the second idea in my project.
For internal and external noise, I followed a similar approach to what happens in a differential amplifier, which is to take signals from two locations and sum them together until the noise cancels each other out.
However, I didn't use a differential amplifier in the system because it requires one of the signals to be inverted. This is difficult to do without changing the gain, so I used a method of summing the electrical signals using resistors.

After analyzing the ECG signal file using FFT on MATLAB [Figure 2.2], it was found that the ECG signal falls in the low frequency range.
So I took that into consideration in my design of the amplifier so that it would not cut off the signal.

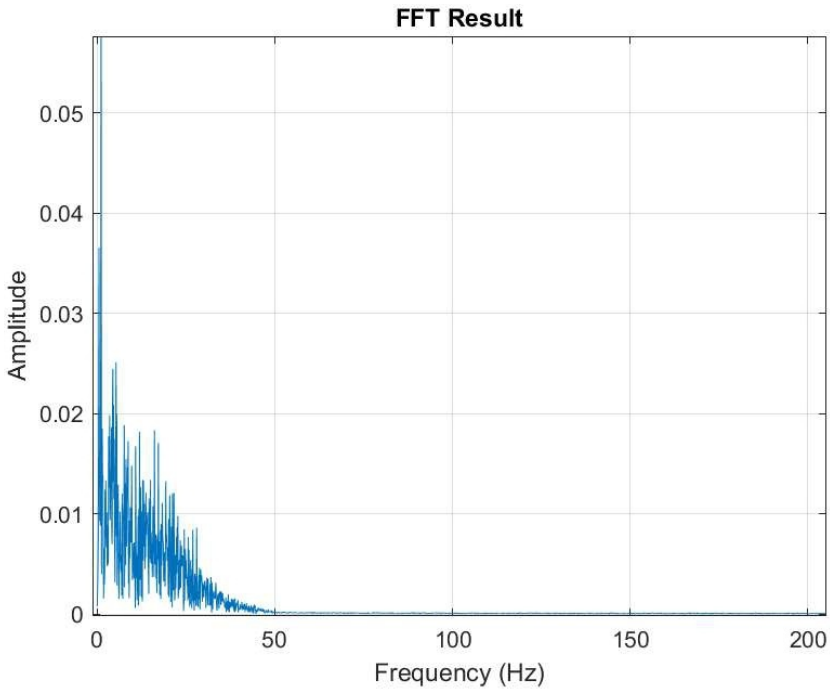

Figure 2.2: FFT of a 10 sec sample ECG signal

After amplifying the two signals and combining them with resistors, I used a clipper circuit that cuts the signal after it exceeds 0.7 volts.
However, I replaced the clipper circuit with a transistor switch circuit because I only wanted a pulse representing the passage of the R signal, not the R signal itself.

Now that I have a pulse that represents the passage of the R wave, I need to amplify this small pulse, enlarge its width, and display it on something that can display it.
To amplify this pulse, I used a hybrid circuit consisting of a voltage buffer, a transistor switch, and a

common emmitter amplifier. This allowed me to clarify the pulse, amplify it, and increase its current.

To display it, I placed a buzzer at the end of the system so that it would make a beeping sound whenever a pulse crossed it.
To prolong the pulse duration, I used a capacitor parallel to the buzzer so that it would not turn on and off too quickly, so that we could notice the buzzer sound.

3 Implementation

Three ECG/EEG Electrodes [Figure 3.1] are connected to the right thenar eminence, the left thenar eminence, and the right elbow. On the other side, they are connected to crocodile wires directly in the circuit. The electrodes connected to the thenar eminences were used as signal inputs. The third electrode, connected to the elbow, was connected to the system ground so that the input signals would not be floating points.

After the signals were entered into the system, they were passed through a voltage buffer. A JFET transistor was chosen because of its high input resistance. The signal was then passed through two common emitters for amplification. After amplification, the two signals were combined using two resistors connected at the same point and to the output terminals resulting from the two amplified signals.

This results in an amplified signal with only the R pulse above 0.7 volts. The signal will then pass through a simple switch circuit consisting of a transistor and a resistor, which will cut out everything below 0.7 volts, which is noise and useless signals.

The switch circuit sometimes shows a portion of the T wave, so I conducted several experiments to remove this wave. What I found was that the amplifier circuit's lower cutoff voltage does not include the range in which this wave is located. Therefore, I have amplified the pulse and removed the unwanted signal.

Then I made the signal pass again through the switch circuit to make it purer and larger. After the switch circuit, I added a voltage buffer circuit like the one at the beginning of the system to produce a signal that could turn on the buzzer.

Because the signal is so fast, it's difficult to hear. I placed a capacitor in parallel with the buzzer, which has an operating voltage of 5 volts. Because the pulse input to the buzzer has a voltage greater than 5 volts, the buzzer remains active until the capacitor's charge drops below 5 volts. Therefore, the buzzer's sound can now be heard.

Figure 3.1: Electrodes

The previously explained system was represented by an electrical circuit on a breadboard, as shown in Figure 3.2.
Capacitors were connected between VCC and ground to reduce noise.

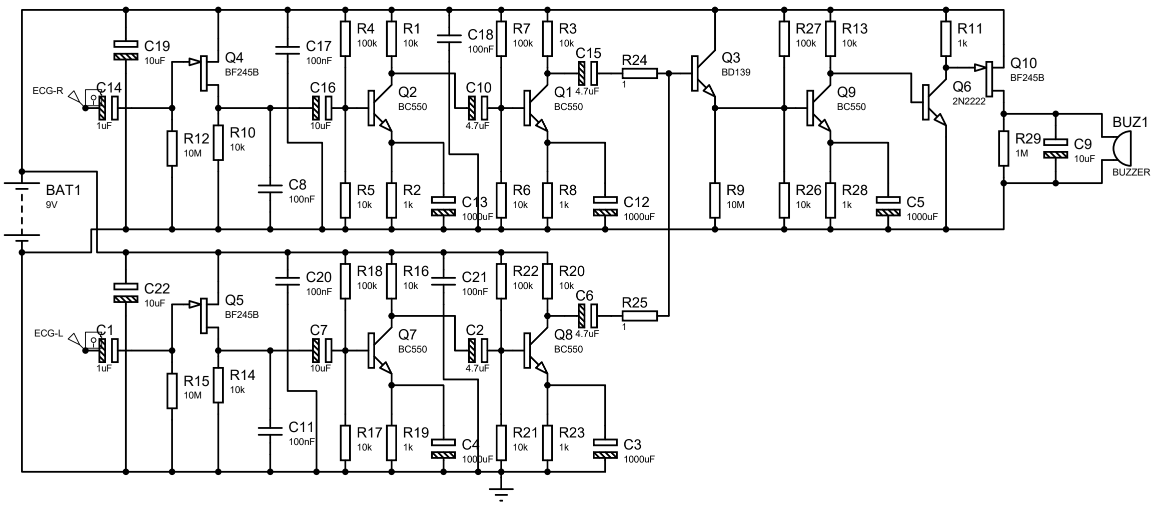

Figure 3.2: Electrodes

4

Analyses & calculations

After designing the system, I analyzed it using Proutuse and LTspice.
And because my system isn't just a multistage amplifier circuit, analyzing the AC isn't meaningful. When analyzing the frequency response of the two identical amplifiers, each one gave this frequency response.

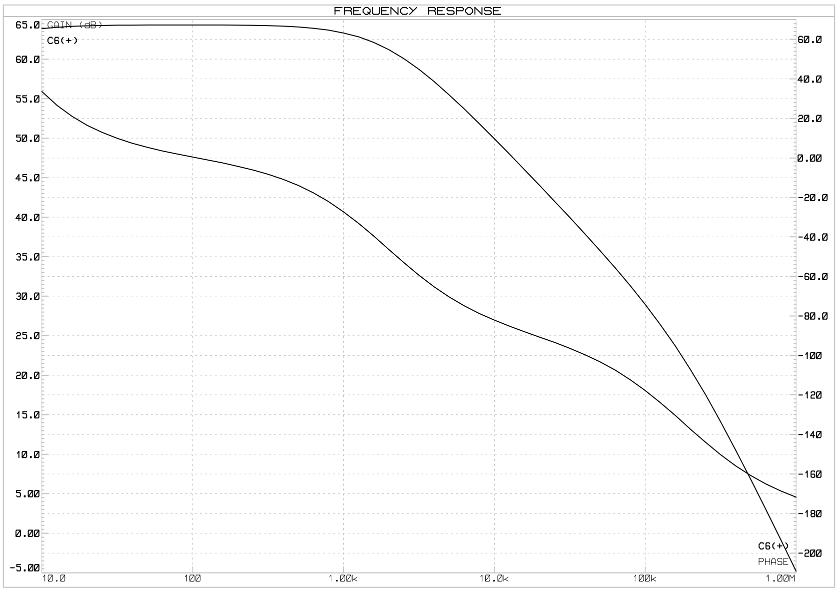

Figure 4.1: FFT of a 10 sec sample ECG signal

5

Conclusion

This project took a lot of time and effort, especially since I was responsible for implementing it alone.
It was also physically exhausting, as I had to travel to downtown Cairo several times, as the circuit was modified approximately six times.
The idea of the project failing, which repeatedly crossed my mind, was the biggest obstacle to my progress.
I even considered canceling it and replacing it with a regular audio amplifier project.
However, in the final days before the deadline, I was able to get the project up and running, and the excitement I felt that day is indescribable.
This project will never be forgotten.